

SIMIONESCU ROBERT-ALEXANDRU

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Summary

Computer Science student at the Technical University of Cluj-Napoca with hands-on experience in FPGA/digital design (5-stage pipelined MIPS CPU) and full-stack web development (React, Django, FastAPI), including production sites delivered for real clients.

Education

Technical University of Cluj-Napoca

Computer Science – C.T.I (Faculty of Automation and Computer Science)

Cluj-Napoca, Romania

Oct. 2024 – Present

- 3rd Year Undergraduate Student
- **Relevant coursework:** Object-Oriented Programming, Computer Architecture, Operating Systems, Databases, Data Structures & Algorithms, Digital System Design
- **Current semester:** Software Engineering, Structure of computer systems, Graphic processing, Design with microprocessors

"Andrei Muresanu" National College

Mathematics and Computer Science Specialization

Bistrita, Romania

Sep. 2020 – Jul. 2024

- Baccalaureate Exam Grade: **9.00 / 10**

Experience & Volunteering

Atomic Bots FIRST Tech Challenge Team

2022 – Present

Co-founder, Programming branch (Alumni, still active)

Bistrita, Romania

- Co-founded the Atomic Bots robotics team and coordinated the technical and programming effort for FIRST Tech Challenge competitions.
- Designed and implemented autonomous navigation and control algorithms in **Java** (Android Studio), integrating multiple sensors.
- Built the team's web interface from scratch and optimized its UI/UX.
- Continue to support the team after graduating: maintaining the website and mentoring new members.

Projects

Veres Coffee and Bakery Website | *HTML, CSS, JavaScript, Cloudflare Pages*

Jul. 2026 – Sep. 2026

- Designed and built a responsive presentation website for a local business, with an intuitive menu section and simple site navigation.
- Deployed on **Cloudflare Pages** with continuous deployment from a **Git** repository: every push is built and published automatically.
- Reached **1.4k unique visitors** and **19.9k requests** in the first month after launch, with **65%** of traffic served from Cloudflare's edge cache.

Calorie Tracker Bot | *JavaScript, Cloudflare Workers, Telegram Bot API, Google Gemini*

2026

- Built a serverless **Telegram bot** on **Cloudflare Workers** that logs meals and estimates calories and macros with the **Gemini API**.
- Created a custom Android widget with progress rings for daily calories, carbs and water intake; used daily.

32-bit MIPS Microprocessor | *VHDL, FPGA (Nexys A7), Xilinx Vivado, MIPS ASM* Mar. 2026 – May 2026

- Designed and implemented a fully functional 32-bit **MIPS** processor from scratch, with both Single-Cycle and **5-Stage Pipelined** architectures in one repository.
- Engineered the datapath and control units and handled data and control hazards through static resolution (compiler-level NOP insertion, branch delay slots) and inter-stage pipeline registers.
- Validated the CPU with custom MIPS assembly programs (e.g., mathematical computations), simulated with Vivado testbenches and run on the FPGA with output on the 7-segment display.

"U" Engine | *React, Next.js, Python, FastAPI, Google Gemini*

Apr. 2026

- Developed a full-stack tactical analysis platform that aggregates and processes **Wyscout** football data through a custom **Python/FastAPI** backend.
- Built an AI module with the Google GenAI SDK (Gemini 2.5 Flash), using Dynamic Context Injection (opponent profile, coach persona, raw stats) to generate structured **JSON** tactical strategies.
- Designed an interactive **React/Next.js** frontend with dynamic pitch visualizations and statistical radars; reduced API latency and operational costs with a custom in-memory caching layer.

YouTube Downloader GUI | *C++, Qt (In progress)*

Feb. 2026 – Present

- Built a desktop GUI in **C++** and **Qt** as a visual interface for the *yt-dlp* utility.
- Implemented download and processing of YouTube videos and playlists in multiple formats (MP4, MP3, FLAC).

Atomic Bots Web Platform | *React, Styled Components, JavaScript* Jan. 2026

- Designed and implemented the responsive frontend for the robotics team in **React** and **Styled Components**, with a modular component architecture.
- Implemented Single Page Application navigation with **React Router** and prepared the build for static hosting on the institution's servers.

Leadership Coaching Platform | *Python, Django, SQLite, HTML/CSS* Dec. 2025 – Jan. 2026

- Developed a full-stack web platform for a client, including a custom admin panel for independent blog article management.
- Designed the relational database in **SQLite** and built a contact form with automated confirmation emails over **SMTP**.
- Handled deployment and maintenance on **PythonAnywhere**, including **cPanel** administration and **DNS** configuration for the custom domain.

IoT Air Quality Monitoring System | *C++, Arduino UNO R4 WiFi* Dec. 2025

- Designed an environment monitoring system with multiple sensors (DHT11, MQ-2, MQ-135) for temperature, air quality and flammable gas detection.
- Implemented a linear-interpolation algorithm ("Gradient Smooth") for smooth RGB LED color transitions via **PWM**, reflecting temperature changes.
- Integrated **NTP** over WiFi to synchronize time on an **I2C** LCD display, removing the need for an RTC module.

Hardware VGA Controller | *VHDL, FPGA (Nexys A7)* Apr. 2025 – May 2025

- Implemented a hardware **VGA controller** on the **Nexys A7** FPGA that renders geometric shapes in multiple colors, controllable from board switches.
- Designed the control FSMs and counters that generate the **H-Sync** and **V-Sync** signals.
- Built a frequency divider to derive the 25 MHz *pixel clock* from the 100 MHz board clock.

Nintendo Land | *C#, .NET* Apr. 2023 – May 2023

- Developed a desktop virtual game store inspired by Steam, with authentication, dynamic user profiles (avatars, XP leveling), reviews and trailer playback.

Competitions

"U" Hack – FC Universitatea Cluj Cluj-Napoca, Apr. 2026

- Built and pitched the "**U Engine**" prototype in a fast-paced hackathon, combining Wyscout data with AI to give technical staff actionable tactical strategies.

Lovable HoHoHack 2025 Cluj-Napoca, Dec. 2025

- Worked in a team to build an online tutoring prototype using Lovable: studdybuddyfullstackinut.lovable.app

Coderun 2025 Cluj-Napoca, Nov. 2025

- Competed in 5 technical challenges covering React, Python/Django, C++/SDL3, Arduino and cybersecurity.

Technical Skills

Programming Languages: C, C++, C#, Java, Python, JavaScript, TypeScript, VHDL, MIPS Assembly

Web: React, Next.js, Django, FastAPI, HTML/CSS

Hardware & Embedded: FPGA (Nexys A7), Xilinx Vivado, Arduino, PWM, I2C

Desktop: Qt, .NET, Java Swing, SDL3

Databases: SQL, SQLite, MySQL

Tools: Git, GitHub, Linux, Visual Studio Code, IntelliJ, Android Studio, CLion, DataGrip

Spoken Languages: Romanian (Native), English (C1, Cambridge CAE, score 192, Nov. 2023), French (B1), Hungarian (A2), German (A1)

Courses and Certifications

Database Foundation & Programming with SQL – Oracle Academy Feb. 2026

Google Cybersecurity Seminars – Technical University of Cluj-Napoca 2026

CyberSea Festival 2026 – Certificate of Participation 2026

Personal Development

Certified Neuro-Linguistic Programming (NLP) Practitioner – NLP Integrativ Apr. 2026